

Rohan Patil

Software Developer

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OBJECTIVE

Software Developer with M.Tech in Computer Science and Engineering, specializing in backend development, ML deployment, and full-stack solutions. Proven expertise in building production-grade REST APIs, cloud infrastructure (AWS), and scalable systems handling thousands of users. Skilled in Python, FastAPI, PHP, React, and AWS services with hands-on experience in ML model deployment (Detectron2, SageMaker), database architecture, and microservices. Passionate about developing efficient, scalable solutions that drive business value.

WORK EXPERIENCE

Junior Software Developer, *Climatescence Pvt Ltd – Pune, India*September 2025 – Present

- Technologies – Python, FastAPI, PHP, React.js, PostgreSQL, MySQL, AWS (S3, SageMaker, SNS, EC2), REST APIs, Git, GitHub.
- Architected and developed production-grade REST APIs using FastAPI for multiple enterprise applications, serving thousands of users with 99.9% uptime and sub-200ms response times for critical endpoints
- Built scalable backend systems handling large-file uploads (GeoTIFF, orthomosaic, DEM) with async I/O processing, AWS S3 integration, and automated workflow orchestration
- Deployed and optimized GPU-based ML inference services (Detectron2) on AWS SageMaker with auto-scaling (scale-to-zero when idle), reducing infrastructure costs by 40% while maintaining real-time inference capabilities
- Developed comprehensive approval workflow engine supporting multi-step, role-based sequential approvals across payment, grievance, inventory, and operational modules
- Built end-to-end grievance management module with file attachments, status tracking, comment threads, and role level access control serving thousands of farmer records
- Created Farmers Journey tracking system using complex SQL CTEs to fetch farmer details, parcel data, documents, and statuses in single optimized queries, reducing API response time by 60%
- Designed two-dimensional access control system combining roleid and levelid for fine grained permissions across 50+ pages and 200+ actions with reusable permission traits
- Collaborated with cross functional teams including product managers, data scientists, and frontend developers to deliver 12+ major features on time in agile sprints

INTERNSHIPS

Android Developer Intern, *Technolite Info Solutions – Pune, India*January 2023 – June 2023

- Technologies – Android Studio, Java, Firebase.
- Managed android projects implementing key functionalities and deploying time-critical client requests.
- Performed extensive manual testing for app versions.

Python Developer Intern, *Global Engineering Deans Council – India Chapter*June 2021 – August 2021

- Technologies – Python, Jupyter, R, Machine Learning, Deep Learning.
- Managed machine learning and deep learning projects implementing key functionalities.
- Designed and implemented a image detection model using a Convolutional Neural Network.

EDUCATION

Walchand College of Engineering, SangliJune 2025

MTech in Computer Science and Engineering

- CGPA: 8.10/10.0

Rajarambapu Institute of Technology, IslampurJune 2023

BTech in Computer Science and Information Technology

- CGPA: 7.02/10.0

PROJECTS

Human Behavior Analytics Using Surveillance System, Mtech ThesisAugust 2024 - June 2025 🌐

- Technologies: Python, Deep Learning, OpenCV, Flask, React.*
- Developed an AI-driven system leveraging CCTV footage to analyze human behavior patterns.
- Implemented a YOLOv11-based deep learning model for accurate, real-time person detection in CCTV footage. Integrated CNN-based facial expression recognition, achieving 87% accuracy in real-time emotion detection.
- Designed a Flask-based analytics pipeline to track and summarize emotional trends over time. Enabled automated export of emotion data in CSV format for daily, weekly, and monthly analysis.
- Built a responsive, interactive dashboard using React to visualize behavioral insights and analytics.

Agro Nation (Blockchain-Web Application), Capstone ProjectMarch 2022 - Jan. 2023 🌐

- Technologies: Blockchain, React, Solidity, MetaMask.*
- Implemented a web application that maintains supply chain of crops using blockchain technology.
- It facilitates the services to farmers for maintaining the distribution of crops.

Agri Info (Android Application)

Feb. 2022 - June 2022 

- *Technologies:* Android Studio, Java.
- Developed an automated system to validate and verify the farm data and generate the required information to work in the field.
- The android application provides the correct and useful information that will help them to maximize their crop production.

CERTIFICATES

Python Course by Google, Coursera 

Complete Android Course - Java, Udemy 

Blockchain Basics, Coursera 

SKILLS

Programming Languages: Python, FastAPI, PHP, Java, JavaScript, Flask.

Tools / Technologies: Machine Learning, Artificial Integilance, React, Google Colab.

Cloud & DevOps: AWS (S3, SageMaker, SNS, ECR, EC2), Docker, Git, GitHub.

Database System: PostgreSQL, MySQL

Interpersonal Skills: Problem Solving, Teamwork, Leadership, Communication, Time Management, Agile Collaboration.